

CNN

Convolutional Neural Network

visual data process,

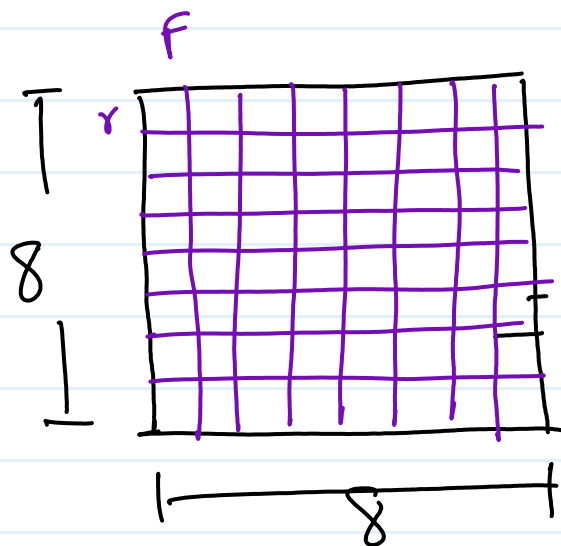
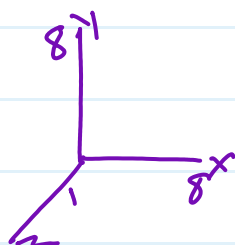
Human brain $\rightarrow$ visual cortex

- stage
- V_1 - orientation, edges,
 - V_2 - texture and complex shapes
 - V_3 - motion and dynamic form
 - V_4 - color and form of object
 - V_5 - motion perception and moving object tracking.

Gray scale -

Black = 0
white = 255

$(8, 8, 1)$

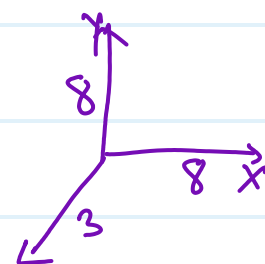
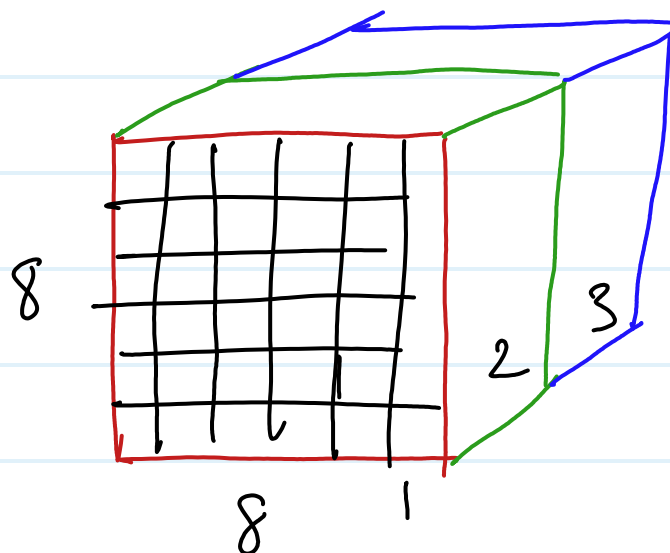


RGB

$$R = 255$$

$$G = 255$$

$$B = 255$$



$(8, 8, 3)$

R □

$(255, 0, 0)$

B □

$(0, 0, 255)$

G □

$(0, 255, 0)$

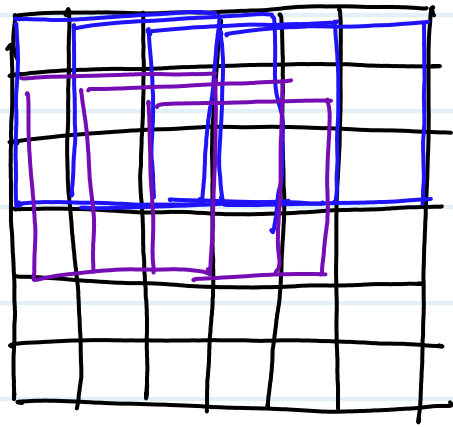
75% RB 25%

$(200, 0, 75)$

3

$[0, 0, 120]$

convolutional operation



Filter / kernel
 3×3
 Horizontal

-1	0	1
-2	0	2
-1	0	1

i/p = 6×6

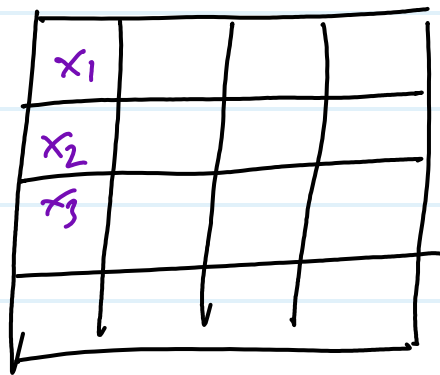
vertical

1	2	1
0	0	0
-1	-2	-1

stride = 1

$$o/p = (n - f) + 1$$

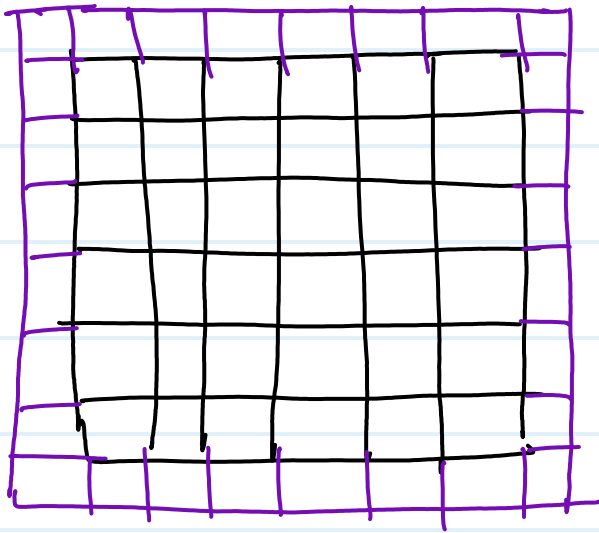
$$= (6 - 3) + 1 \Rightarrow 4$$



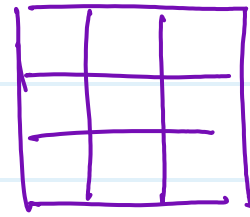
4×4

6x6

Padding - 1



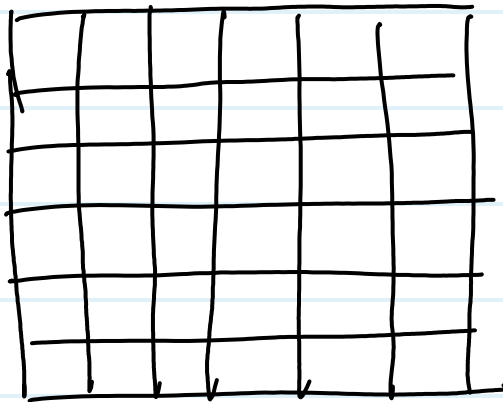
8x8



3x3

$$\Rightarrow (n-f) + 1$$

$$\Rightarrow (8-3) + 1 \Rightarrow 6$$



6x6

To calculate padding size

$$\text{ip } (n-f) + 2P + 1 = 6^{\text{op}}$$

$$(6-3) + 2 \times P + 1 = 6$$

$$3 + 2P + 1 = 6$$

$$4 + 2P = 6$$

$$2P = 6 - 4$$

$$P = \frac{2}{2} = 1$$

Pooling / Subsampling

① max

② min

③ Avg.

① max

4	3
5	4

② min

0	0
0	0

③ Avg

1.75	1.5
2.25	1.75

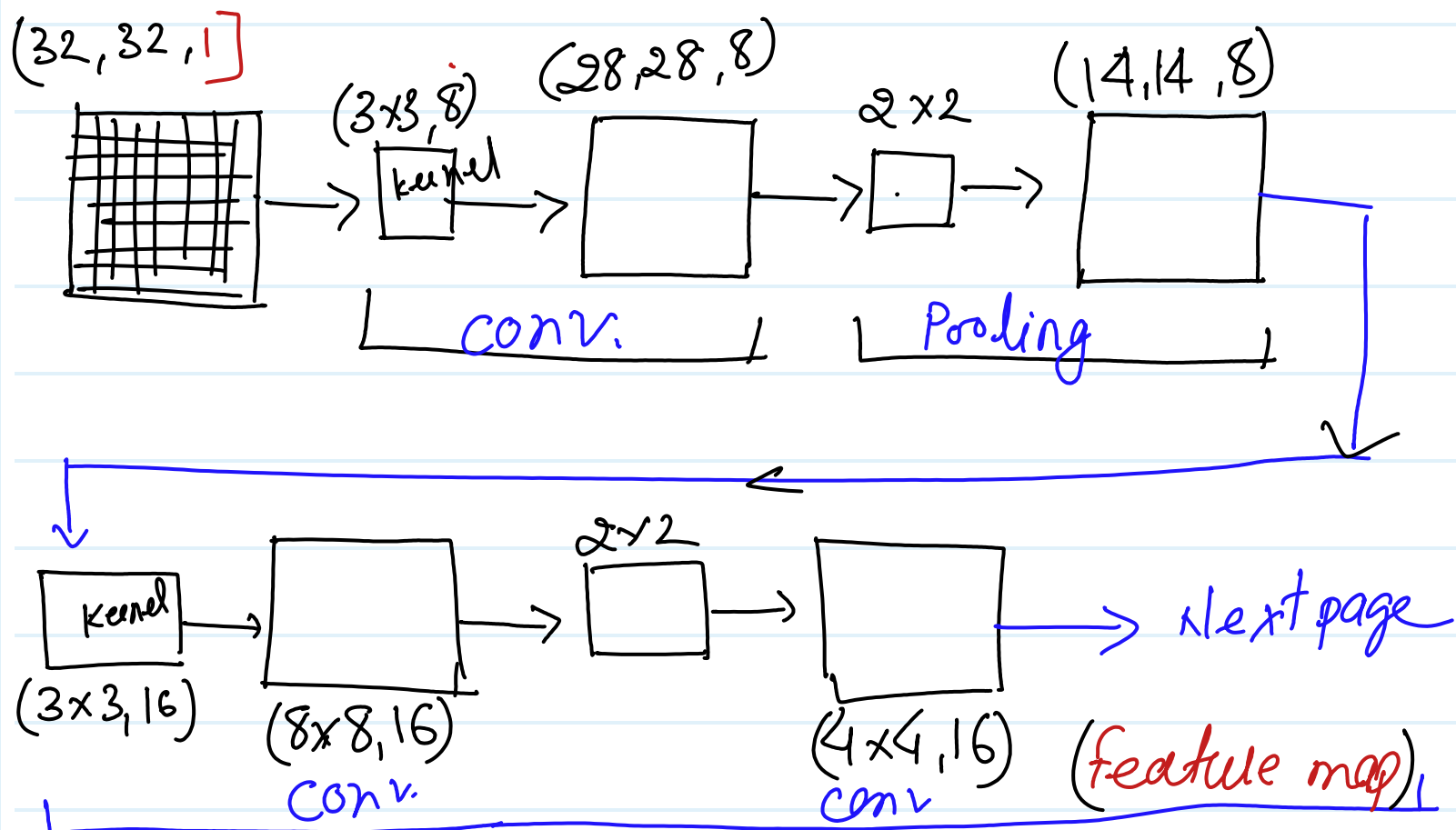
4

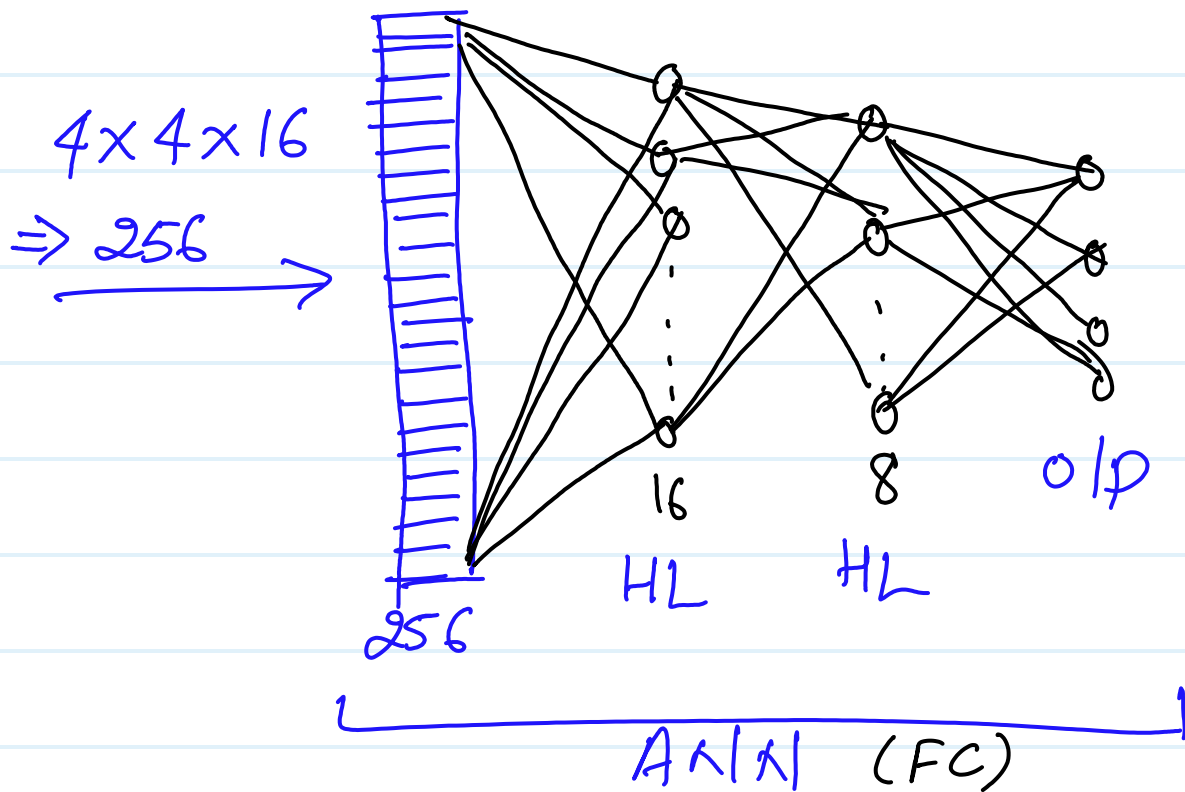
1	2	2	0
0	4	1	3
0	1	0	4
5	3	2	1

4

CNN

- ① conv. layer
- ② kernel / filter
- ③ stride
- ④ Padding
- ⑤ Pooling
- ⑥ Flatten
- ⑦ fully conn. layer
- ⑧ Dropout
- ⑨ Batch normalization





feature map + ANN = CNN

famous CNN Architecture

- ① LeNet-5
- ② Alexnet
- ③ VGG-16
- ④ VGG-19
- ⑤ Inception (Google)
- ⑥ Resnet (Microsoft)
- ⑦ EfficientNet

→

0	100	10	220
65	50	60	1
180	80	120	25
20	135	255	89

↑

$$\frac{10}{255} = 0.04$$

$$\frac{200}{255} = 0.8$$

$$\frac{255}{255} = 1$$

$$\frac{0}{255} = 0$$

(0 - 1)

	f_1	f_2	f_3
↑	.	.	.
.	.	.	.
.	.	.	.
.	.	.	.
.	.	.	.

